

Tailored Virtual Agent Guidance for Stress Management: Comparing Directive and Non-Directive Approaches by Alexithymia Status

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Abstract

Stress significantly affects young adults, highlighting the need for effective digital mental health interventions (DMHIs). However, many DMHIs face high dropout rates and low engagement, partly due to alexithymia, characterized by difficulties in identifying and expressing emotions. Directive guidance support individuals with alexithymia, however, DMHIs rely on one-size-fits-all strategies. This study investigated a large language model (LLM)-powered virtual agent delivering Directive or Non-Directive guidance for stress management and compared two guidance styles according to alexithymia status. The results showed that although stress reduction did not differ significantly across the two guidance styles, individuals with alexithymia reported enhanced therapeutic alliance under Directive guidance, while those without alexithymia formed stronger alliances with Non-Directive guidance, particularly for tasks ($p = .03$) and goals ($p = .05$). The study advances understanding tailored guidance by integrating LLM-powered virtual agent, demonstrating that personalizing interventions for alexithymia can strengthen therapeutic alliance and boost long-term engagement in DMHI.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → *Psychology*.

Keywords

Mental Health, Large Language Model, Virtual Agent

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1 Introduction

Stress is a growing concern in modern societies, particularly among young adults aged from 18 to 29, with academic stress identified as a significant contributing factor [25]. A 2023 survey conducted in the United States found that 79% of young adults reported experiencing stress in the past month, highlighting the urgent need for effective mental health interventions [1]. Left unaddressed, stress significantly increases the risk of mental health disorders, emphasizing the importance of accessible and effective support systems [35].

Digital mental health interventions (DMHIs) have emerged as promising tools, offering evidence-based therapies through mobile applications and online platforms [14]. DMHI facilitates timely interventions for stress management, offering high accessibility and lower costs compared to traditional face-to-face therapy. Among various therapeutic techniques, cognitive restructuring, a commonly used technique in DMHIs, has proven effective in helping individuals reframe negative thoughts into constructive perspectives [30]. However, despite their potential, DMHIs often face challenges such as high dropout rates or low engagement, diminishing their therapeutic impact [9].

One critical factor affecting engagement in DMHIs is alexithymia, a cognitive-emotional processing deficit characterized by difficulties in identifying and articulating emotions [26]. This condition is particularly prevalent among young adults and poses significant challenges for interventions that require introspection and reflection on negative thoughts [23]. For instance, cognitive restructuring often employs on conversations with virtual agents (VAs) that mimic the appearance and interaction of a human therapist, with prompts such as "What evidence supports your negative thought?" [17, 20, 34] While effective for some, this non-directive approach can be overwhelming for individuals with alexithymia, leading to reduced therapeutic alliance and lower intervention efficacy [22, 36].

Recent advancements in artificial intelligence offer promising solutions to these challenges. Large language models (LLMs), as demonstrated by Sharma and colleagues [32], can provide directive guidance by offering explicit advice and reframed perspectives. This

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directive approach effectively reduces the cognitive and emotional burdens associated with interventions like cognitive restructuring. However, it may also limit opportunities for users to independently develop cognitive restructuring skills, a key therapeutic goal of non-directive guidance [18].

To improve DMHIs, it is essential to tailor guidance styles to individual needs, particularly for individuals with alexithymia. Research indicates that directive guidance is particularly beneficial for individuals with alexithymia, as it helps them overcome emotional and cognitive barriers [5]. Young adults with alexithymia require tailored support to address their unique challenges, such as difficulties in recognizing and processing negative emotions [29]. Studies further suggest that providing direct examples and advice regarding emotions and thoughts enhances both therapeutic effectiveness and alliance for individuals with alexithymia, aligning closely with the benefits of directive approaches [27]. Despite these findings, most existing DMHIs rely on uniform, one-size-fits-all strategies, such as psychoeducational videos, which fail to account for individual characteristics [6].

To address these limitations, this study investigates how young adults with and without alexithymia respond to two distinct guidance styles (Directive vs. Non-Directive) in DMHIs. We developed VAs powered by LLMs to deliver stress interventions, each incorporating one of these guidance styles. Using a 2×2 factorial design, participants interacted with both Directive and Non-Directive VAs, enabling direct comparison of their experience. Semi-structured interviews were also conducted to explore the influence of alexithymia on intervention outcomes. The study is guided by three primary objectives:

- Develop LLM-powered VAs incorporating Directive and Non-Directive guidance styles
- Evaluate stress reduction and therapeutic alliance formation among young adults with or without alexithymia in response to the two guidance styles
- Examine the influence of alexithymia on stress reduction and therapeutic alliance through qualitative analysis (i.e., semi-structured interviews)

2 Methods

2.1 Participants

This study recruited 22 participants from a university community in Seoul, Republic of Korea between December 2024 and January 2025, including 10 participants identified as having alexithymia and 12 not having alexithymia. Alexithymia was assessed using a validated questionnaire administered after recruitment (see Section 2.2). Participants were selected based on specific inclusion criteria: (1) they were aged between 18 and 29, consistent with the defined range for young adults [2], (2) they experienced unresolved stress episodes related to academic situations, as the intervention targeted academic stress to minimize the variability in stress causes, and (3) they were fluent in Korean, as the experiment was conducted entirely in Korean.

Exclusion criteria were applied to exclude individuals diagnosed with psychiatric disorders such as depression or anxiety, ensuring that external psychiatric characteristics did not influence the study outcomes. All participants provided written informed consent after

receiving a detailed explanation of the study procedures, including the use of VAs for research purposes. The study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki and was approved by the Institutional Review Board of Seoul National University of Science and Technology.

2.2 Alexithymia Scale

Participants were classified as having or not having alexithymia using the Korean version of the Toronto Alexithymia Scale (TAS-20)[4]. The TAS-20 is a widely recognized and highly reliable tool for identifying individuals with alexithymic, with a Cronbach's alpha of 0.85[12]. Higher scores on the TAS-20 indicate greater difficulty in identifying and expressing emotions.

The scale comprises 20 items rated on a 5-point Likert scale and evaluates three key dimensions of alexithymia: Difficulty Identifying Feelings (DIF), Difficulty Describing Feelings (DDF), and Externally Oriented Thinking (EOT). All participants completed the TAS-20, and their total scores were used to classify them into one of two groups based on established criteria: individuals scoring above 61 were categorized as Alexithymia, while those scoring below 60 were classified as Non-Alexithymia.

2.3 Stress Intervention with VAs

The stress intervention employed a conversation with a VA using Unreal Engine 5.3, guided by cognitive restructuring principles. The intervention was delivered to participants on a laptop (Intel i7-10700, 32 GB RAM, and NVIDIA GeForce RTX 3070) in a private room, separate from the researcher. Participants provide input to the VA via the laptop's keyboard, and the VA automatically generates responses based on GPT-4o. The VA displayed empathetic facial expressions in response to the participant's input and delivered messages through chat bubbles and synthesized speech.

The intervention followed a structured framework consisting of four stages, according to study by Shidara et al. [34]: (1) Opening Talk, which engaged in small talk to build a therapeutic alliance and gathered contextual information about situations, emotions, and negative thoughts of participants; (2) Evaluating Thought, which examined the validity and fsepotential impact of negative thought; (3) Reframing Thought, which focused on transitioning negative thoughts with more rational perspectives; and (4) Closing Talk, which summarizes the session and reinforces key insights for future application. Both Directive and Non-Directive VAs followed this framework while employing distinct guidance styles to accomplish each stage.

Directive VA analyzed responses and provided advice using a multi-agent system, inspired by the study of Sharma et al. [32]. This system comprised three agents: a thinking trap classifier to identify cognitive distortions, a reframing thought generator to propose alternative perspectives, and a conversation orchestrator to deliver the generated information in an empathetic manner, as if based on personal experience of VA. For instance, in the second stage, the Directive VA provided personalized advice, such as, "I have also thought, 'I am not as good as others.' However, this is like comparing your worst to someone else's best," instead of advising, "You're caught in a thinking trap known as 'comparing and despairing.'" In the third stage, the Directive VA proposed alternative

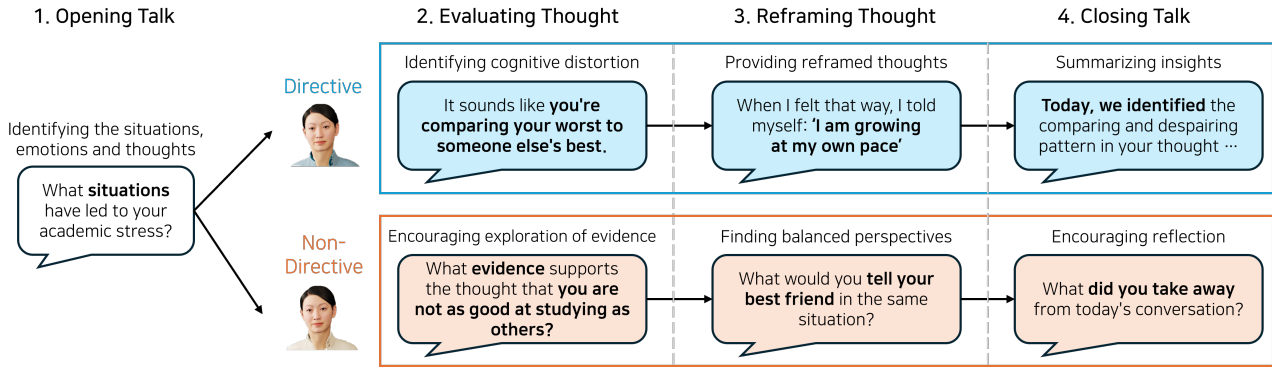


Figure 1: Example of interactions with Directive and Non-Directive VAs during a stress intervention. The Directive VA provides direct guidance and restructured perspectives to aid cognitive restructuring, while Non-Directive encourages self-reflection through open-ended questions.

perspectives to help participants reframe their negative thoughts into constructive ones. Finally, in the Closing Talk, the Directive VA summarized the key insights discussed during the intervention. This approach allowed the Directive VA to deliver direct guidance and concrete suggestions tailored to each participant's context.

In contrast, Non-Directive VA offered open-ended questions without direct advice. These questions were derived from cognitive restructuring script for VA by Shidara [33] and modified by the LLM, called conversation generator, to fit participants' situations. For instance, in the second stage, the Non-Directive VA asked participants to evaluate their thoughts by requesting evidence, such as, "What evidence supports the thought that you are not as good at studying as others?" This encouraged participants to assess the rationality of their beliefs. In the third stage, the nondirective VA guided participants to consider potential future scenarios and adopt alternative perspectives. The intervention concluded with participants summarizing insights they could apply beyond the session, enabling them to explore cognitive patterns at their own pace.

2.4 Questionnaire

Two questionnaires were utilized to assess participants' stress levels and the therapeutic alliance formed with the VA. First, stress level was measured using a 100-point Likert scale (0 = "Not stressful," 100 = "Panic"), adapted from a daily stress inventory [10]. The scale was administered to compare pre-intervention and post-intervention stress levels between two guidance styles. Second, the therapeutic alliance with the VA was evaluated by Working Alliance Inventory–Short Revised (WAI-SR)[16]. This 12-item questionnaire, rated on a 5-point Likert scale (1 = "Strongly disagree," 5 = "Strongly agree"), measures three dimensions of alliance—agreement on goals, agreement on tasks, and interpersonal bond—using four items per dimension. The WAI-SR was translated into Korean, and "therapist" was replaced with "virtual therapist"[7].

2.5 Procedure

All participants experienced both Directive and Non-Directive interventions, based on a Latin-square design. The experiment began with the completion of the TAS-20, which assessed alexithymia in

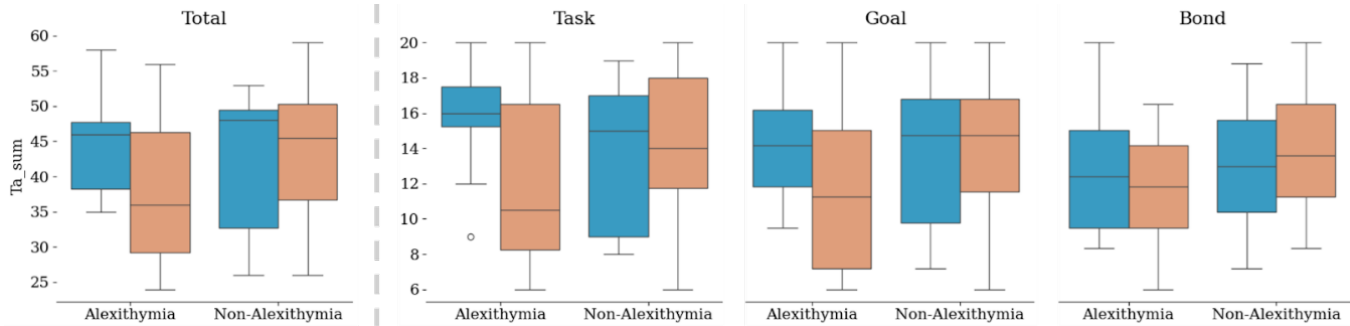
about five minutes. Next, participants filled out a pre-intervention stress questionnaire and engaged in the stress intervention for an average of 12 ± 5 minutes, focusing on the issue that caused their stress. Immediately afterward, they completed a post-intervention stress questionnaire to measure changes in their stress levels. After a short break of up to five minutes, the same procedure was repeated with the other guidance style. Participants addressed a new stress episode in the pre-intervention stress questionnaire, engaged with the VA intervention, and completed the post-intervention stress questionnaire. Upon completing both interventions, participants were asked to complete the WAI-SR for each VA. We then conducted semi-structured interviews to compare the two guidance styles, focusing on which VA was more helpful. The entire procedure lasted approximately 50 minutes.

2.6 Analysis

To evaluate the effectiveness of the guidance style across alexithymia status, statistical analyses were conducted using Python 3.10. Chi-squared tests and independent-samples t-tests were used to compare demographic characteristics and alexithymia scale results between Alexithymia and Non-Alexithymia. And a linear mixed-effects model (LMM) was applied to analyze stress level and therapeutic alliance. LMMs were chosen as they account for potential effects caused by individual variability. For stress level analysis, the LMM included the main effects of time (i.e., pre- and post-intervention), alexithymia status, and guidance style, as well as the interaction effect between alexithymia status and guidance style. To analyze the therapeutic alliance, the LMM were conducted, including the main effects and interaction effects on the alexithymia status and guidance style. Additionally, thematic analysis was conducted on the semi-structured interview to explore participants' experiences of the guidance style and the influence of alexithymia on their experiences.

Table 1: Comparative analysis of basic demographic characteristics and TAS-20 results between Alexithymia and Non-Alexithymia

	Alexithymia (N=10)	Non-Alexithymia (N=12)	p-value
Basic Demographic Characteristics			
Gender (female)	5	6	> .99
Age (years)	23.00 ± 1.38	22.17 ± 1.55	.07
TAS-20 Results			
Total Score	60.30 ± 9.70	46.75 ± 3.82	< .001
DIF	19.40 ± 6.18	14.42 ± 3.67	.002
DDF	18.10 ± 3.96	11.25 ± 2.10	< .001
EOT	22.80 ± 4.76	21.08 ± 3.35	.17

**Figure 2: Boxplots comparing the WAI-SR scores for Directive (blue, left) and Non-Directive (orange, right) interventions across Total, Task, Goal, and Bond between Alexithymia and Non-Alexithymia.**

3 Results

3.1 Basic Demographic Characteristics and Alexithymia Scale Results

As shown in Table 1, chi-squared and independent-samples t-tests revealed no significant differences in basic demographic characteristics between the Alexithymia and Non-Alexithymia groups. However, an independent-samples t-test showed significant differences in the TAS-20 total score, DIF, and DDF, while no difference was found in the EOT. This result is due to the relatively low internal consistency of the EOT on the TAS-20 compared to the DIF and DDF [21]. Nevertheless, following previous studies confirming that the TAS-20 total score is a reliable measure for categorizing alexithymia [28], participants were classified into Alexithymia and Non-Alexithymia groups based on their TAS-20 total score.

3.2 Stress Level Change Result

The results of LMM revealed a significant main effect of time ($\beta = 16.78, p < .001$), indicating that stress levels decreased from pre- to post-intervention. A significant main effect of alexithymia status was also found ($\beta = 17.01, p = .002$), with participants with alexithymia reporting higher stress levels compared to Non-Alexithymia. This result is consistent with previous studies that have identified alexithymia as a predictor of increased stress [15, 24]. Conversely, the main effect of the guidance style ($\beta = 4.08, p = .33$)

and the interaction effect between alexithymia status and guidance style was not statistically significant ($\beta = -4.08, p = .50$).

3.3 Therapeutic Alliance Results

The results of LMM showed the main effects of alexithymia status were not significant across all therapeutic alliance scores, including the total score, task, goal, and bond. Similarly, the main effects of the guidance style were not significant for the total score, task, goal, and bond. However, significant interaction effects between alexithymia status and guidance style were observed for the total score ($\beta = 8.30, p = .03$), task ($\beta = 3.98, p = .03$), and goal ($\beta = 2.82, p = .05$), while the bond was not statistically significant ($\beta = 1.50, p = .13$; detailed in Table 4 in the Appendices). These results indicate that therapeutic alliances, except for the bond subscale, were stronger when Alexithymia interacted with the Directive VA and Non-Alexithymia interacted with the Non-Directive VA.

3.4 Semi-structured Interview Results

Thematic analysis of the interviews revealed that participants compared the helpfulness of the two guidance styles in terms of cognitive and emotional support. As summarized in Table 2, Alexithymia reported positive cognitive and emotional support from the Directive Guidance, while Non-Alexithymia reported positive cognitive and emotional support from the Non-Directive Guidance.

Alexithymia reported that Directive guidance was more effective for stress management. Directive guidance helped Alexithymia shift

Table 2: Thematic analysis of semi-structured interviews with alexithymia and non-alexithymia participants interacting with Directive and Non-Directive VAs. Positive experiences are indicated by (+) and negative experiences by (-).

Alexithymia status	Guidance Style	Cognitive Support	Emotional Support
Alexithymia	Directive	(+) The VA presented new perspectives, encouraging participants to imagine alternative ways of thinking.	(+) The intervention with VA was neither burdensome nor difficult, allowing participants to engage comfortably.
Alexithymia	Non-Directive	(-) The intervention was limited to summarizing their thoughts, making it challenging for participants to discover new perspectives.	(-) Participants found the VA's questions frustrating and overlapping, making the process of reflecting on their situations uncomfortable.
Non-Alexithymia	Directive	(-) The VA's advice was perceived as superficial and formulaic, failing to provide participants with meaningful insights.	(-) The intervention lacked sufficient depth to explore their emotions and situations thoroughly, resulting in no significant impact.
Non-Alexithymia	Non-Directive	(+) The VA encouraged participants to think more deeply, enabling them to consider their situations from a different perspective.	(+) Participants felt a sense of trust and reassurance because the intervention resembled their existing stress regulation process.

negative thinking to a more constructive perspective by providing advice to emulate. In contrast, Non-Directive guidance lacked clear direction, making it difficult for these participants to reflect on their negative thoughts. Alexithymia also described Non-Directive guidance as frustrating and difficult to interpret, and some participants reported experiencing emotional burden when asked to reflect on their negative feelings. These findings align with previous research, indicating that individuals with Alexithymia may struggle with cognitive restructuring [8]. Overall, the results suggest that guidance styles offering clear direction and positive modeling help people with alexithymia manage stress more effectively.

In contrast, Non-Alexithymia evaluated Non-Directive guidance as more effective for stress management. Non-Directive guidance's cognitive support allowed participants to engage in deeper reflection and view their stressful situation from new perspectives, thereby facilitating meaningful cognitive restructuring. Directive guidance, however, was perceived as superficial and formulaic, offering little substantial impact on cognitive change. Furthermore, Non-Alexithymia noted that Non-Directive guidance felt more familiar and comfortable, likely because they had prior experience with self-directed cognitive restructuring. Overall, the results suggest that a guidance style encouraging self-exploration helps individuals without alexithymia manage stress more effectively.

4 Discussion

4.1 Principal findings

The study highlights the importance of tailoring guidance styles based on the alexithymia status to enhance the therapeutic alliance in DMHIs. We developed an LLM-powered VA incorporating cognitive restructuring techniques and two distinct guidance styles: Directive and Non-Directive. The study hypothesized that the effectiveness of these guidance styles would vary depending on the alexithymia status of participants.

While the results did not reveal significant differences in stress reduction based on alignment between alexithymia status and guidance style, notable differences were observed in the therapeutic alliance. Specifically, significant variations were found in the total score and in the Task and Goal subdimensions of WAI-SR (i.e., therapeutic alliance). These findings suggest that Directive guidance may be more effective for individuals with Alexithymia, while Non-Directive guidance may better support those without Non-Alexithymia in fostering a strong therapeutic alliance.

Contrary to our hypothesis, tailored guidance style did not result in greater stress reduction. Two potential explanations for this outcome are worth noting. First, both Directive and Non-Directive guidance were derived from well-established interventions known to effectively reduce stress. As both interventions demonstrated statistically significant stress reduction, the differences between them may not have been pronounced. Second, the study was conducted as a single-session experiment, which may have limited the ability to capture the cumulative effects of tailored guidance. Previous research suggests that tailored guidance styles often rely on sustained engagement and the gradual strengthening of the therapeutic alliance [3]. Future studies should explore the long-term effects of tailored guidance styles, building on this study's insights into fostering initial therapeutic alliance formation.

The therapeutic alliance findings align with our hypothesis, demonstrating that guidance styles tailored to alexithymia status significantly enhance the Task and Goal subdimensions of the alliance. WAI-SR results revealed that participants perceived the tailored guidance as more collaborative and effective. For instance, a participant with alexithymia (P1) noted that the Directive guidance provided clear advice they were ready to act on. In contrast, the same participant expressed confusion after interacting with the Non-Directive VA, stating, "I don't know what I'm supposed to do." These results align with prior research, which highlights the benefits of concrete advice and examples for individuals with

alexithymia [27, 29], suggesting that this approach is also effective in VA-driven interventions.

Interestingly, there were no significant differences in the Bond subdimension of the therapeutic alliance, which appears to be influenced more by nonverbal factors. Over half of the participants noted feeling understood due to the VA's facial expressions, which were implemented identically across both VAs. This finding supports previous research by Choi et al. [11], which identified nonverbal expressions as core factors for conveying empathy in VAs. However, disagreements on goals and tasks can lead to alliance ruptures, even when the bond remains intact [13], highlighting the importance of tailoring guidance styles to participants' alexithymia status.

This study demonstrates that tailoring guidance styles to alexithymia status enhances the therapeutic alliance in DMHIs. While the use of LLMs tailored to personal characteristics has been explored in domains such as education [31] and technology [19], its application in DMHIs remains underexplored. To the best of our knowledge, this study is the first to implement an LLM-powered VA to provide tailored guidance for individuals with alexithymia, offering valuable insights into advancing tailored digital interventions. Although the findings did not confirm that tailored guidance styles significantly enhance stress reduction, they suggest that a strong therapeutic alliance may provide additional therapeutic benefits over the long term. This study provides initial evidence that tailoring interventions to personal characteristics enhances the therapeutic alliance in DMHIs, offering a foundation for future research into personalized digital interventions.

4.2 Limitation

This study has several limitations that should be considered when interpreting the findings. First, the small sample size limits the generalizability of the results. Future research should include larger and more diverse participant groups to enhance the robustness and applicability of the findings across different populations. Second, the study did not investigate the long-term effects of the intervention, leaving uncertainties about the sustained impact of tailored guidance styles on therapeutic outcomes. Longitudinal studies are needed to evaluate how tailored guidance styles influence therapeutic alliance and outcomes over time, particularly in real-world, multi-session interventions. Finally, this study focused exclusively on alexithymia as a personal characteristic, raising questions about whether the observed effects can be replicated with other individual characteristics. Future research should explore the applicability of tailored guidance styles to a broader range of personal traits, such as cognitive styles, emotional regulation capabilities, or cultural differences, to improve the generalizability and versatility of tailored interventions.

4.3 Conclusion

This study provides preliminary evidence that tailoring guidance styles in VAs to individuals' alexithymia status can significantly enhance the therapeutic alliance in DMHIs based on cognitive restructuring. By comparing Directive and Non-Directive guidance styles, the findings revealed that both approaches were effective in reducing stress levels, regardless of alexithymia status. However,

individuals with alexithymia experienced greater benefits from Directive guidance, while those without alexithymia formed stronger therapeutic alliances with Non-Directive guidance, particularly in the Task and Goal components of the alliance. These results underscore the importance of designing tailored DMHIs that account for personal characteristics to foster a stronger therapeutic alliance, which is a critical factor in the success of interventions. Future research should explore the long-term effects of tailored guidance styles on therapeutic alliance formation and stress reduction, as well as extend this approach to other personal characteristics to enhance the scalability and efficacy of DMHIs.

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A Questionnaire

A.1 Alexithymia Scale

Please read each statement and indicate how much you agree or disagree with the statement (5-point Likert Scale).

- (1) I am often confused about what emotion I am feeling.
- (2) It is difficult for me to find the right words for my feelings.
- (3) I have physical sensations that even doctors don't understand.
- (4) I can describe my feelings easily.
- (5) I prefer to analyze problems rather than just describe them.
- (6) When I am upset, I don't know if I am sad, frightened, or angry.
- (7) I am often puzzled by sensations in my body.
- (8) I prefer to just let things happen rather than to understand why they turned out that way.
- (9) I have feelings that I can't quite identify.
- (10) Being in touch with emotions is essential.
- (11) I find it hard to describe how I feel about people.
- (12) People tell me to describe my feelings more.
- (13) I don't know what's going on inside me.
- (14) I often don't know why I am angry.
- (15) I prefer talking to people about their daily activities rather than their feelings.
- (16) I prefer to watch "light" entertainment shows rather than psychological dramas.
- (17) It is difficult for me to reveal my innermost feelings, even to close friends.
- (18) I can feel close to someone, even in moments of silence.
- (19) I find examination of my feelings useful in solving personal problems.
- (20) Looking for hidden meanings in movies or plays distracts me from my enjoyment.

A.2 Stress Level Questionnaire

A.2.1 Pre-intervention.

- (1) Have you experienced a situation related to academic pressure that caused you stress, and the issue remains unresolved? Please recall and describe the stressful situation in detail. Write at least 3 sentences.
- (2) While reflecting on the situation described above, how intense is the stress you currently feel? Please rate your current stress intensity on a scale from 0 (not stressful) to 100 (panic).

A.2.2 Post-intervention.

- (1) After your conversation with the virtual therapist, how intense is the stress you currently feel regarding the situation? Please rate your current stress intensity on a scale from 0 (no stress) to 100 (panic).

A.3 Therapeutic Alliance Questionnaire

Rate the statements regarding your experience talking to the virtual therapist from strongly disagree to strongly agree (7-point Likert scale).

- (1) I have a clearer understanding of how I can change after the conversation. (*Task*)

- (2) The conversation gave me a new perspective on my problems. (*Task*)
- (3) I believe the virtual therapist likes me. (*Bond*)
- (4) The virtual therapist and I collaborated to set the goals of the conversation. (*Goal*)
- (5) The virtual therapist and I respect each other. (*Bond*)
- (6) The virtual therapist and I worked toward mutually agreed-upon goals. (*Goal*)
- (7) I feel welcomed by the virtual therapist. (*Bond*)
- (8) The virtual therapist and I agree on what is important to me. (*Goal*)
- (9) I feel that the virtual therapist cares for me even if I don't follow their instructions. (*Bond*)
- (10) The content of the conversation will help me achieve the changes I want. (*Task*)
- (11) The virtual therapist and I have a good understanding of the beneficial changes I need. (*Goal*)
- (12) I believe the way I am addressing my problems is correct. (*Task*)

(Note: All italicized texts do not appear in the version of the questionnaire that we provided for the participants.)

B Detailed Results

B.1 Stress Level

Table 3 presents detailed results from a linear mixed-effects model analysis examining the effects of time, alexithymia status and guidance style on stress levels. Coefficient values are calculated with guidance set to "Directive" and alexithymia status set to "Alexithymia". Time is calculated with "pre-intervention". Coefficient values are calculated with guidance set to "Directive" and alexithymia status set to "Alexithymia". Time is calculated with "pre-intervention".

B.2 Therapeutic alliance

Table 4 presents detailed results from a linear mixed-effects model analysis examining the effects of alexithymia status and guidance style on therapeutic alliance. Coefficient values are calculated with guidance set to "Directive" and alexithymia status set to "Alexithymia". Coefficient values are calculated with guidance set to "Directive" and alexithymia status set to "Alexithymia". Time is calculated with "pre-intervention".

Table 3: The linear mixed-effects model analysis of the effects of time, alexithymia status and guidance style on stress level.

	β	SE	z	p-value
(Intercept)	32.97	4.09	8.07	< .001
Time	16.78	3.02	5.56	< .001
Alexithymia status	17.01	5.49	3.10	.005
Guidance style	4.08	4.18	0.98	0.31
Alexithymia status x Guidance style	4.07	6.05	-0.68	0.79

Table 4: The linear mixed-effects model analysis of the effects of alexithymia status and guidance style on therapeutic alliance.

	β	SE	z	p-value
Total				
(Intercept)	43.50	2.89	15.06	< .001
Alexithymia status	-5.70	4.28	-1.33	.18
Guidance style	-1.50	2.53	-0.59	.55
Alexithymia status x Guidance style	8.30	3.76	2.21	.03
Task				
(Intercept)	14.17	1.23	11.48	< .001
Alexithymia status	-1.967	1.83	-1.08	.28
Guidance style	-0.58	1.21	-0.48	.63
Alexithymia status x Guidance style	3.98	1.81	2.21	.03
Goal				
(Intercept)	14.83	1.11	13.43	< .001
Alexithymia status	-2.03	1.64	-1.24	.22
Guidance style	-0.42	0.99	-0.42	.67
Alexithymia status x Guidance style	2.82	1.46	1.93	.05
Bond				
(Intercept)	14.50	0.90	16.17	< .001
Alexithymia status	-1.70	1.33	-1.28	.20
Guidance style	-0.50	0.67	-0.75	.45
Alexithymia status x Guidance style	1.50	0.99	1.52	.13